

Kurtosis

how heavy are the tails
how extreme are the outliers

- | | | |
|-----------------------|--------------------------------------|-------------------------------------|
| ① Mesokurtic | ② Leptokurtic | ③ Platykurtic |
| → Normal distribution | Tall peak
+ heavy tails | flat distribution |
| excess kurtosis = 0 | many outliers
excess kurtosis > 0 | few outliers
excess kurtosis < 0 |

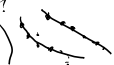
Q-Q plot Quantile-Quantile plot

{ Actual quantiles vs theoretical quantiles }

Is my data normally distributed?

Interpretation rule

- If points lie on diagonal line the data is normal
- If points deviate significantly, data is not normal



Residual plots

↳ Regression models

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

- If residuals are random: Good model
- If residual shows pattern: Model assumptions violated